

S5 — Leakage Insights Manager & MQTT Isolation (with live telemetry)

OpenAMI / MeshEMS — Continuous Leakage Detection (NESL)

Allahaddje Bonheur — IoT Fellow, 10Power (in collaboration with NESL)

2026-07-27

1 Objective

Sprint **S5**: implement the **Leakage Insights Manager** and the **multi-EMS nomination** that isolates a leak on a shared LV feeder, grounded in the literature (`S5_literature_findings.md`), and prove it end to end — including **live leakage telemetry over MQTT** on a real network.

Result: S5 is complete. The manager + nomination run on the ESP32; the leakage streams over MQTT to a broker on the PC, fault flagged, and is **charted live**.

2 Design — nomination (magnitude + peer-correlation)

Each EMS publishes leakage telemetry; the round ranks candidates by **magnitude** *and* by the “**odd one out**” (the node whose ramp signature correlates least with its peers), then a **confidence gate** decides auto-isolate (bracket upstream+downstream) vs alert-only.

S5 nomination — magnitude + peer-correlation, confidence-gated (literature-grounded)

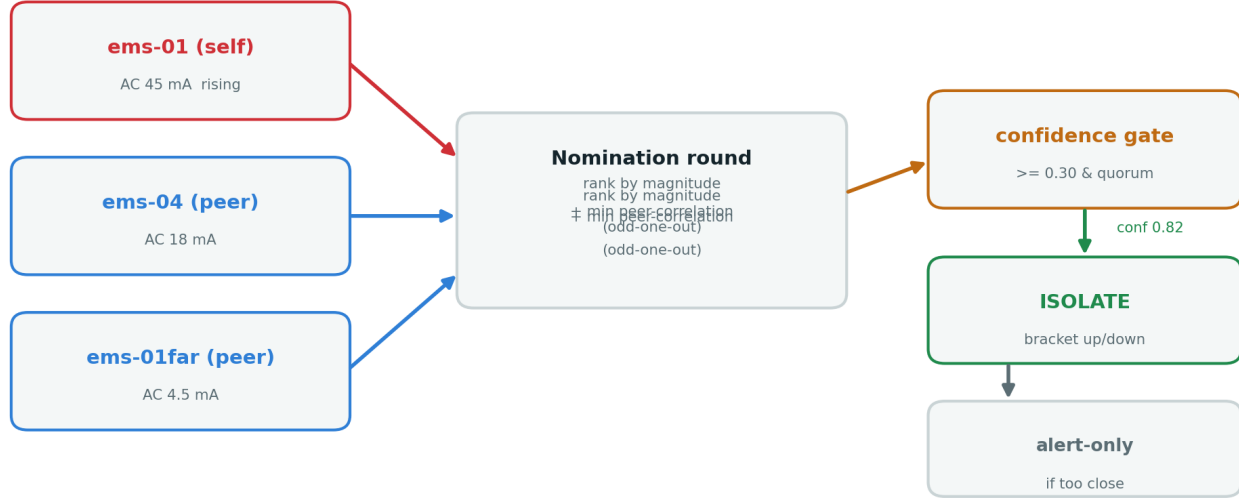


Figure 1: S5 nomination flow — magnitude + peer-correlation, confidence-gated.

3 Implementation

include/metering/leakage_insights_manager.h + .cpp:

- **LeakageInsightsManager** — per-EMS state machine (normal → watch → isolating → fault → restored), telemetry, onset + ramp, and a ramp **signature** for correlation.
- **nominate()** — same phase + **correlated onset** → rank by magnitude → **min-peer-correlation** odd-one-out → **confidence** (boosted when magnitude leader == min-correlation node, de-rated under reverse power flow) → **isolate** (0.30 + quorum) or **alert-only**.
- **selfTelemetry** / **roundJson** / **alertJson** — the three MQTT payloads.

4 Evidence A — 3-EMS nomination (bench mock)

This EMS (mock ramp) + two simulated peers (mid 0.4×, far 0.1×):

```

S5 isolation: {"nominated_ems":"street-ems-01","rule":"max_ac_mA + min_peer_corr, correlated_ons
              "confidence":0.82,"action":"isolate_takeoff (bracket upstream+downstream)",
              "reason":"mag+corr agree","quorum":{"reporting":3}}
  
```

The nominated node is both the magnitude leader **and** the min-correlation (odd-one-out) node → mag+corr agree → confidence **0.82** → isolate. The full round runs on **one ESP32**.

5 Evidence B — MQTT over the real network

The ESP32 connects to WiFi (a phone hotspot) and to **Mosquitto on the PC**; the PC subscriber receives the leakage, fault included:

```
openami/StreetEMS_<id>/subpanel_RCMleaks
{"acSinusoidal":{"value_mA":30.66,"threshold_mA":30,"inFault":true},
 "dc":{"value_mA":6.13,"threshold_mA":6,"inFault":true}, ...}
```

(A **WPA3** / **PMF** hotspot blocked the ESP32's older WiFi stack; a **WPA2** hotspot connected instantly.)

6 Evidence C — live charts (MQTT Explorer)

Plain-number topics `rcm_ac_mA` / `rcm_dc_mA` are published so any dashboard can chart the leakage. The mock ramp is visible **rising then holding** at the ceiling (AC → 45 mA, DC → 9 mA):

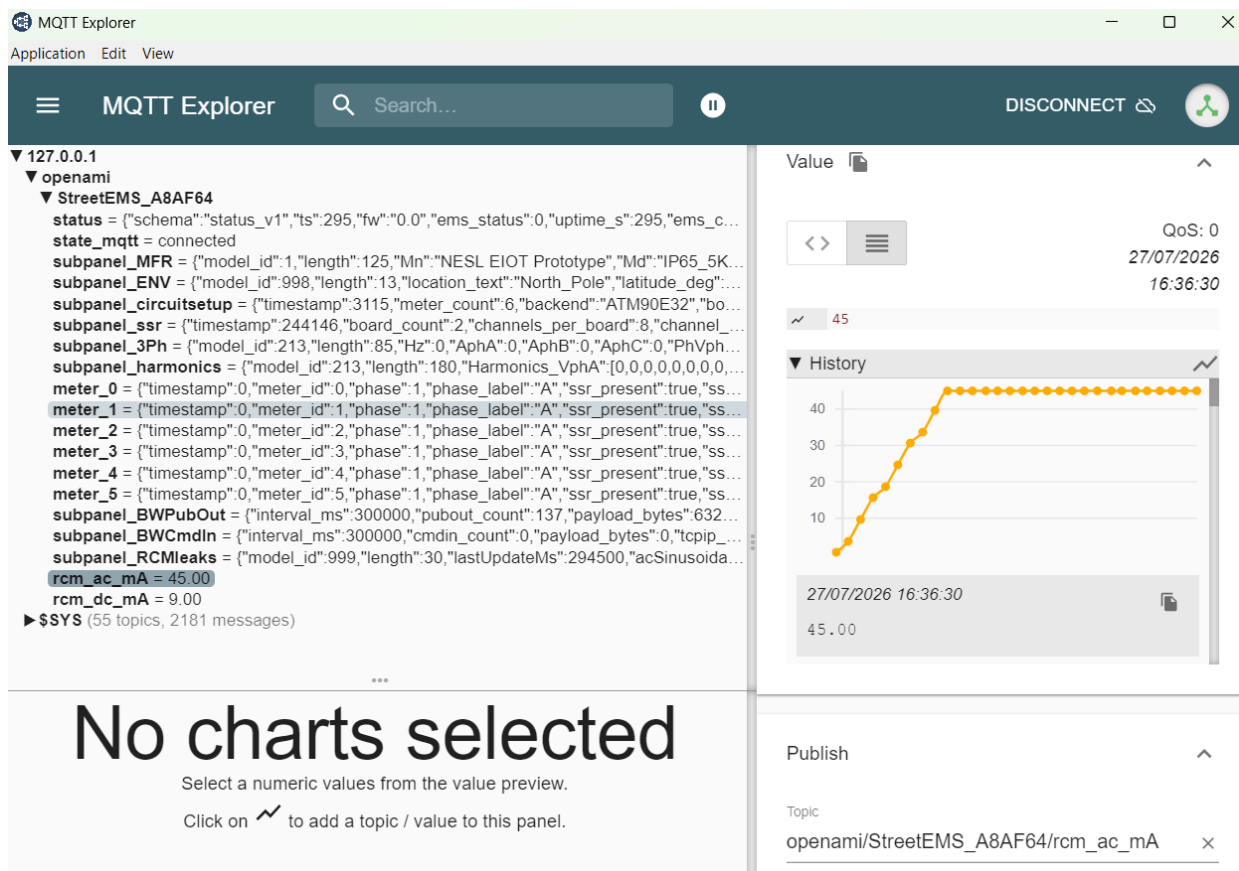


Figure 2: AC leakage — `rcm_ac_mA` rising to 45 mA then holding (MQTT Explorer live chart).

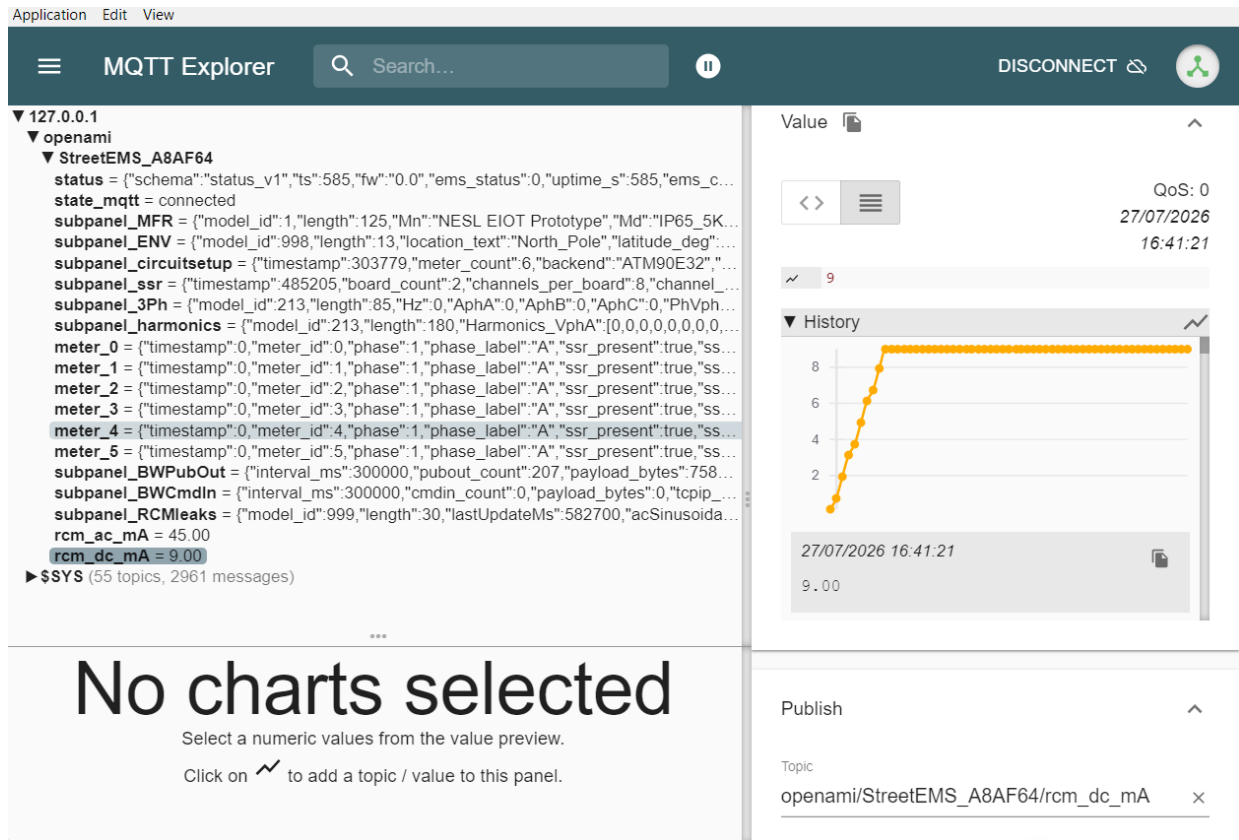


Figure 3: DC leakage — `rcm_dc_mA` rising to 9 mA then holding (MQTT Explorer live chart).

7 Literature grounding (see S5_literature_findings.md)

- **Nomination by min-correlation** (“odd one out”), not raw amplitude — robust to EMS at different feeder positions (from the faulty-feeder-ID method, read in full).
- **Direction** by the residual-voltage / residual-current phase angle $|\mathbf{V}_{\text{res}} - \mathbf{I}_{\text{res}} - 90^\circ|$ — the reserved `phase_angle_deg` field / IVY-upgrade path.
- **Bracketed isolation** (both sides) + auto-vs-operator mode — from FLISR practice.
- **Change-based** trigger (not absolute level).

8 Status & next

- **S5 complete** — insights + nomination (literature-grounded), proven on-device; leakage telemetry live over MQTT with fault flag and live charts.
- Production polish: publish the S5 telemetry/isolation/alert on the shared `lvfeeder/...` topics and subscribe to **real peers** (replace the simulated ones).
- **Next: S6** — hardware alarm-line GPIO + reverse-power-flow threshold (27 mA).

Repository: docs/leakage_MD0630TA1A/ — details in S5_implementation.md, S5_literature_findings.md, and the interactive demo S5_isolation_demo.html. Companion to S1–S4 reports; together they form the full technical document.